



Bioconversion of *Terminalia arjuna* bark powder into a herbal feed for *Labeo rohita*: Can it be a sustainability paradigm for Green Fish production?

D.K. Meena^a, A.K. Sahoo^a, M. Jayant^b, N.P. Sahu^b, P.P. Srivastava^{b,c}, H.S. Swain^a, B.K. Behera^a, K. Satvik^a, B.K. Das^{a,*}

^a ICAR-Central Inland Fisheries Research Institute, Barrackpore, Kolkata 700120, West Bengal, India

^b ICAR-Central Institute of Fisheries Education, Mumbai 400061, India

^c Dr. Rajendra Prasad Central Agriculture University, Samastipur 848125, Bihar, India

ARTICLE INFO

Keywords:

Carcass composition

Growth indices

Nucleic acid

Nutrient retention

Labeo rohita

Terminalia arjuna bark powder

ABSTRACT

Surging demand of fish production for nutritional and livelihood security manifested with irrational use of synthetic growth enhancer and immunostimulants which are fostering the awful health concerned to human being and negatively affecting the beneficial micro-biota in aquatic ecosystem. The need of hour is to use environment friendly herbal therapy, as an alternate to those chemicals and compounds. The *Terminalia arjuna* is known for its excellent array of ethnomedicinal properties including antimicrobial, antioxidant and was reported to have positive effects on metabolomics of the animals. A 90-day experiment was conducted to ensure effective valorization of *Terminalia arjuna* bark powder (TABP) based feed in *Labeo rohita*. Four iso-nitrogenous {305.74 g crude protein (CP)/ kg diet} and iso-caloric {18.15 MJ gross energy (GE)/kg diet} experimental diets were formulated incorporating 0.0, 5.00, 10.00 and 15.0 g TABP /kg feed and designated as TABP0, TABP5, TABP10 and TABP15, respectively. 540 juveniles (average weight, 20.7 g, 45 fish per tank) were randomly stocked in 12 fibre reinforced plastic (FRP) tank (500 L water capacity). Fish were fed twice at 9.30 h and 5.00 h to satiation level. Results of this study revealed the highest growth rates {weight gain percentage (WGP) and specific growth rate (SGR)}, nutrient utilization {protein efficiency ratio (PER), apparent net protein utilization (ANPU), fat retention (FR) and energy retention (ER)} and survival percentage (SRP) in *L. rohita* at 10 g TABP/ kg feed. While feed conversion ratio (FCR) followed an inverse trend ($P < 0.05$) to growth rates. Growth rates, feed conversion, and nutrient utilization were similar in TABP0 and TABP15 group ($P > 0.05$). Viscerosomatic index (VSI) and craniosomatic index (CSI) were significantly affected ($P < 0.05$) while hepatosomatic index (HSI) was insignificant ($P > 0.05$). Whole body composition was influenced ($P < 0.05$) excluding ash content ($P > 0.05$). RNA content and RNA: DNA ratio were positively correlated to growth rates and nutrient

Abbreviation: ADFom, acid detergent fibre (exclusive of residual ash); nADFom, neutral detergent fibre assayed with a heat stable amylase (exclusive of residual ash); ANPU, apparent net protein utilization; AOAC, association of official analytical chemists; APHA, American public health association; AWG g, average weight gain; BHT, butylated hydroxytoluene; CP, crude protein; CSI, craniosomatic index; EE, ether extract; ER, energy retention; FAO, food and agricultural organization; FCR, feed conversion ratio; FI, feed intake; FR, fat retention; FTI, fat intake; GE, gross energy; HSI, hepatosomatic index; PER, protein efficiency ratio; PI, protein intake; SGR, specific growth rate; TABP, *Terminalia arjuna* bark powder; VSI, viscerosomatic index; WG, weight gain; WGP, weight gain percentage.

* Corresponding author.

E-mail address: basantakumard@gmail.com (B.K. Das).

<https://doi.org/10.1016/j.anifeedsci.2021.115132>

Received 29 July 2020; Received in revised form 21 October 2021; Accepted 22 October 2021

Available online 27 October 2021

0377-8401/© 2021 Elsevier B.V. All rights reserved.

retentions ($P < 0.05$). Thus, present study advocated that TABP at 12.3 g/kg could be utilized efficiently to produce healthy fish for human consumption thereby called Green Fish. Also, paved a prolific way to produce an effective TABP based medicated feed on large scale, having tried in field conditions individually and with commercial feed.

1. Introduction

The fish protein is considered as easily digestible and cheaper as compared to other animal protein (FAO, 2016). The fish, *Labeo rohita* play a vital role in six species composite aquaculture in South East Asian countries including India, due to its compatibility with other species and consumer preferences and contribute major chunk of total fish production. The current fish production is not sufficient to meet the surging demand as world per capita fish consumption is around 20.3 kg (FAO, 2020). Every step towards the intensification of culture practice for enhancing fish production is provoking numerous stresses in the culture environments that lead to disease and mass mortality. The conventional approach of treatment of disease is antibiotics and irrational use of chemicals and substances for growth promoter and immunity booster have alarmed the risk of developing disease resistant strains of bacteria and residual of obnoxious chemicals in the fish flesh. Antibiotics residues in fish flesh are matter of great concern for human consumption. In addition, antibiotics not only act on the pathogenic bacteria but also affect beneficial and mineralizing micro-biota; leading to loss of beneficial flora, those are responsible for harbouring and recycling the useful nutrients from sediments. These limitations created a quest among the researchers to find out an alternate, which may be cheaper, easily applicable along with health beneficial effects.

Bibliographic studies showed that some researchers have worked for evaluating the effects of herbal materials as growth promoter, immunity booster, stress reliever and a health supplements in fish including *L. rohita* (Shakya, 2017). Further, many studies have proved that herbal additives enhance the growth of fishes and protect from diseases (Rao et al., 2006; Sahu et al., 2007a, 2007b, 2008). In addition, growth of a fish is a function of feed and its feed supplements in the form of prebiotic, probiotic and attractant (De et al., 2014; Hassan et al., 2014). Medicinal plants are blessed with secondary metabolite that are responsible for bio-efficacy, however at higher concentration they interfere normal functional physiology of digestion, assimilation of feed. Some new terminologies have been coined for designating either healthy fish or associated products such as Green growth, Green labels, Green fish and Aquaponics (Dutta, 2015). Now a days a concept of producing the healthy fish, free from obnoxious traces of chemicals applying herbal material called “Green fish” in the line of organic crop products, has come into picture and constantly popularizing across the globe. This type of product getting momentum due to their health benefits and harnesses the medicinal advantages of ethno-medicinal properties in the form of dietary products.

The Arjuna, *Terminalia arjuna* is a plant abundantly found in India. The tree bark of Arjuna is used medicinally in human for numerous health beneficial effects. Amalraj and Gopi (2016) have recorded the beneficial medicinal properties of *T. arjuna* such as antioxidant, hypotensive, antiatherogenic, anti-inflammatory, anticarcinogenic, antimutagenic and gastro productive effect in human being. However, a comparative analysis of this plant extracts on fish health is very poorly undertaken and the detailed information is scanty. An attempt was made by Meena et al. (2021a) for evaluating the antioxidant properties and free radical scavenging activity of facile matrix extract matrix of *T. arjuna*. Recently, Meena et al. (2021b) reported bio-efficacy of *T. arjuna* solvent extracts against elite microbes of aquaculture importance including bacterial isolates and fungus, *Aphanomyces invadans*. Therefore, the present study was conceptualized to evaluate the *T. arjuna* bark based herbal feed in *L. rohita* for producing Green Fish.

2. Material and methods

2.1. *T. arjuna* bark powder (TABP) or herbal powder production process

Bark of the Arjuna plant was collected from the trees located in ICAR-CIFRI, Campus, Kolkata, India (Authentication no. 14) and confirmed for the botanical origin with the help of morphological characters and botanist (Meena et al., 2020a). Processing involves, washing of the extraneous material, and then drying under sun till removal of the moisture followed by blending in the mixer and sieved through 50 μ m sieve.

2.1.1. Assessment of anti-nutritional factor

Tannin content was estimated following spectrophotometric method of Makkar et al. (1993) with slight modifications. Briefly, a total of 100 μ l of solvent extract for each sample was put in test tube and volume was adjusted to 2.5 ml with DW. Subsequently, 800 ml Folin Ciocalteu was mixed properly. Thereafter, 2.5 ml, 20% Na_2CO_3 solution was mixed followed by 40 min incubation at room temperature. The spectrophotometric reading was noted at 725 nm and concentration was assessed from the standard curve. Saponin was estimated following the method of Harborne (2005) with the following equation. and phytates by Darambazar (2018). In brief, it is a spectrophotometric method that accounts the phytate available in the solvent extracts. Basically, it is based on discoloration of a ferric – sulfosalicylic acid complex as the phytate in the sample replaces the sulfosalicylic acid. A standard curve drawn and concentrations plotted on Y and their absorbance on X for standard curve. Thus phytate expressed in percent that converted in g/kg by applying rationalization of percentage = 10 g/kg.

Saponin (g/kg) = $\text{Wt}_2 - \text{Wt}_1 \times 1000/\text{Wt}_1$.

Where, Wt. = weight of sample, Wt₁ = weight of empty container, Wt₂ = weight of container + extract.

2.2. Diet preparation

Chemical composition of different basic ingredients used to formulate four iso-nitrogenous {305.74 g crude protein (CP)/ kg diet} and iso-caloric {18.15 MJ gross energy (GE)/kg diet} experimental diets was presented in Table 1. Graded levels of TABP 0, 5, 10 and 15 g were incorporated in per kg experimental diets and denoted as TABP0, TABP5, TABP10, and TABP15, respectively as shown in Table 2.

As per the formulation, the required quantity of the feed ingredients were weighed and mixed for homogenous blending followed by cooking in a pressure cooker at 15 psi for 15 min. On cooling of the mixed ingredient, the rest of the ingredients such as fish oil, sunflower oil, butylated hydroxytoluene (BHT, antioxidants), choline chloride (feed attractants), vitamins-minerals mixture and TABP were added to the dough and ensured proper mixing. The mixed ingredients then pressed through a pelletizer having 1 mm die to get uniform sized pellets (1.0 mm diameter). Pellets were collected in aluminium tray and kept in rotating hot air oven at 60 °C for 14 hrs for homogenous drying. Thus, dried pellets were collected in airtight polythene bags and stored at 4 °C until further use.

2.3. Fish maintenance and experimental design

700 fish juveniles of *L. rohita* were purchased from local fish market (average weight, 20.7 ± 0.34 g) and reared in a 5000 L cement cistern for two weeks for acclimatization at ICAR-Central Inland Fisheries Research Institute, Barrackpore, Kolkata, India. A 90-day experiment was conducted in triplicates following a completely randomized design (CRD). Five hundred and forty fish (average weight, 20.7 ± 0.34 g) were distributed into 4 dietary treatments in triplicates. Fish were reared in 500 L tank of a flow-through system and fed twice daily (9.30 am and 5.00 pm) up to satiation. Water quality parameters were monitored on regular basis (APHA, 1995) and maintained in a favourable range viz. temperature, 26.2–27.1 °C; pH, 7.1–7.8; dissolved oxygen, 5.3–6.8 mg/L; total hardness, 210–225 mg/L CaCO₃; ammonia-nitrogen (NH₃-N), 0.12–0.22 mg/L; nitrite-nitrogen (NO₂-N), 0.001–0.004 mg/L; nitrate-nitrogen (NO₃-N), 0.01–0.05 mg/L (Huynh et al., 2016). Monthly sampling was performed to check growth and health status of fish.

After commencement of feeding trial, fish were weighed to assess the growth indices. In the supplementary file the schematic representation of parameters studied has been provided. Final sampling was carried out to determine different response parameters. Three fish from each replicate were sampled randomly and anesthetized by using 0.015 ml/L clove oil and then, sacrificed to assess the body composition, hepatosomatic index (HSI), viscerosomatic index (VSI), and craniosomatic index (CSI). Fish were dissected to collect the tissues from vital organs i.e. liver, kidney gill, intestine and brain in 80% alcohol to assess the DNA-RNA ratio. Fish were starved for 24 h prior to sampling.

2.4. Growth performance, feed conversion, nutrient utilization, body indices and survival

The growth performances, feed conversion, nutrient utilization, body indices and survival were determined by following the standard formulae as given below

- Weight gain percentage (WGP) = {(Final weight-Initial weight)/(Initial weight)} × 100
- Specific growth rate (SGR) = {(Log_e Final weight-Log_e Initial weight)/(Number of days)} × 100
- Feed conversion ratio (FCR) = Feed intake (dry weight)/Body weight gain (wet weight)
- Protein efficiency ratio (PER) = Net weight gain (wet weight)/Protein fed

Table 1

Proximate composition of basal ingredients (dry matter basis).

Nutrients (g/kg)	Fish meal	Groundnut oilcake	Soybean meal	Mustard oilcake	De-oiled rice bran	Wheat flour	TABP ^a
Dry matter (as fed)	932.5	921.3	916.3	913.5	937.5	938.6	937.7
Crude protein	560.0	377.1	451.5	355.8	183.4	121.4	35.4
Ether extract	118.9	84.3	14.8	88.0	6.4	7.7	10.7
Total carbohydrate	165.0	474.1	448.9	489.3	743.9	743.9	767.7
Neutral detergent fibre (aNDFom)	57.1	176.0	148.8	278.9	296.6	111.4	310.2
Acid detergent fibre (ADFom)	6.0	111.2	75.4	187.9	175.2	31.3	186.3
Total ash	156.1	64.5	84.8	66.9	66.3	103.2	281.0
Gross energy ^b (MJ/kg)	20.4	19.0	17.6	18.9	14.5	15.0	9.6
Anti-nutritional factors (g/kg)							
Tannin	–	–	–	–	–	–	7.8 (0.53)
Saponin	–	–	–	–	–	–	5.1 (0.25)
Phytate	–	–	–	–	–	–	1.8 (0.15)

Data expressed as mean (n = 3).

^a TABP – *Terminalia arjuna* bark powder.

^b Gross energy (GE) contents were calculated according to gross caloric values of using the values of 23.6, 39.5, and 17.2 kJ/g for crude protein (CP), ether extract (EE), and total carbohydrate, respectively (Brett, 1973).

Table 2

Feed formulation and chemical composition of the experimental diets.

Ingredients (g/kg, as fed)	TABP0	TABP5	TABP10	TABP15
Fish meal	50.00	50.00	50.00	50.00
Groundnut oilcake	160.00	160.00	160.00	160.00
Soybean meal	280.00	280.00	280.00	280.00
Mustard oilcake	135.00	137.00	140.00	142.00
De-oiled rice bran	145.00	145.00	145.00	145.00
Wheat flour	149.60	142.60	134.60	127.60
Fish oil	30.00	30.00	30.00	30.00
Sunflower oil	30.00	30.00	30.00	30.00
Vitamin mineral mix ^a	20.00	20.00	20.00	20.00
Choline chloride	0.10	0.10	0.10	0.10
Phytase	0.10	0.10	0.10	0.10
Butylated hydroxytoluene	0.20	0.20	0.20	0.20
TABP ^b	0.00	5.00	10.00	15.00
Total	1000.00	1000.00	1000.00	1000.00
Chemical composition of the diets ^c (g/kg) on dry matter basis				
Dry matter (as fed)	923.67	932.70	941.20	945.53
Crude protein	301.90	307.70	304.60	308.77
Ether extract	86.70	87.70	87.83	87.60
Total carbohydrate	533.80	526.43	528.77	523.60
Neutral detergent fibre (aNDFom)	173.87	174.27	174.57	174.76
Acid detergent fibre (ADFom)	95.33	96.33	97.40	98.37
Total ash	77.60	78.17	78.80	80.03
Gross energy ^d (MJ/kg)	18.13	18.18	18.14	18.13
P: E ratio (g protein/ MJ energy)	16.65	16.93	16.79	17.03

^a Composition of vitamin mineral mix (Premix plus) (quantity/kg feed): vitamin A (retinol), 33 mg; vitamin B₂, 40 mg; vitamin B₆, 20 mg; vitamin B₁₂, 0.12 mcg; vitamin D₃ (cholecalciferol), 0.55 mg; vitamin E, 15 mg; vitamin K, 20 mg; calcium pantothenate, 50 mg; choline chloride, 3000 mg; nicotinamide, 200 mg; Co, 9 mg; Cu, 40 mg; Fe, 150 mg; I, 20 mg; Mn, 540 mg; selenium, 2.5 mg; Zn, 100 mg; L-lysine, 200 mg; DL-methionine, 200 mg.

^b TABP – *Terminalia arjuna* bark powder.

^c Chemical composition of the diets – Data expressed as mean (n = 3).

^d Gross energy (GE) contents were calculated according to gross caloric values of using the values of 23.6, 39.5, and 17.2 kJ/g for crude protein (CP), ether extract (EE), and total carbohydrate, respectively (Brett, 1973).

v. Protein retention or Apparent net protein utilization (PR or ANPU) = {(Final carcass protein–Initial carcass protein)/(Protein intake)} × 100

vi. Fat retention (FR) = {(Final carcass fat content–Initial carcass fat content)/(Fat intake)} × 100

vii. Energy retention (ER) = {(Final carcass energy content–Initial carcass energy content)/(Energy intake)} × 100

Body indices were determined as shown below

i. HSI (%) = Liver weight (g)/Weight of fish (g) × 100

ii. VSI (%) = Viscera weight (g)/Weight of fish (g) × 100

iii. CSI (%) = Weight of the brain (g)/(Weight of fish (g) × 100

The survival rate of fish was with following formula

Survival (%) = (Total number of fishes harvested/Total number of fishes stocked) × 100

2.5. Proximate composition of ingredients, experimental diets and fish

Biochemical analysis (proximate composition) of the experimental diets and fish was assessed by adopting the standard methods of the Association of Official Analytical Chemists (AOAC, 1995). For moisture content (AOAC 925.09), samples were dried in a hot-air oven till to achieve a constant weight. Kjeldahl method (AOAC 955.04D) was applied for estimating crude protein (CP) content (N × 6.25). Solvent extraction method (AOAC 945.38 F) was used for assessing ether extract (EE) with an organic solvent (petroleum ether) at 40–60 °C in Soxhlet apparatus (Bhattacharya & Co, Kolkata, West Bengal, India-700004). For estimating total ash content (AOAC 923.03), samples were dried and followed by incineration at 550 °C for 6 hrs in a muffle furnace. Total carbohydrate was determined using following formula-

Total carbohydrate (g/kg) = 1000–{CP (g/kg) + EE (g/kg) + total ash (g/kg)}

Neutral and acid detergent fibres (aNDFom and ADFom) of feed ingredients and experimental diets were determined following the

methods described by Van Soest et al. (1991). Gross energy (GE) contents were calculated according to gross caloric values of using the values of 23.6, 39.5, and 17.2 kJ/g for CP, EE, and total carbohydrate, respectively (Brett, 1973).

2.6. DNA, RNA estimation

The nucleic acid fraction of DNA and RNA in vital tissues was assessed following Pentose analysis method (Schneider, 1957) with slight modification. The concentrations were expressed as µg/100 mg of tissue on dry basis and following equations were drawn (Hassan et al., 2016).

$$\mu\text{g DNA/ml} = (\text{OD at 600 nm})/0.019$$

$$\mu\text{g RNA/ml} = \{(\text{OD at 600 nm} + 0.08) - (\mu\text{g DNA/ml} \times 0.013)\}/0.116$$

2.7. Statistical analysis

Five hundred and forty fish (*L. rohita*) were randomly allocated in 12 experimental tanks (500 L capacity; 45 fish/ tank) in flow-through system to study the effect of graded levels of TABP on different variables. The experiments was conducted under controlled conditions and experimental units (tanks volume, colour, shape and fish weight) were homogenous. The experimental unit was the replicate tank for all determined parameters. Completely randomized design was followed in this study. The effect of dietary TABP graded levels were studied using one-way analysis of variance (ANOVA) in General Linear Model procedure of IBM SPSS Statistics 20.0 (IBM Corp. Released, 2011). The data were tested for homogeneity of variances using Levene test. Polynomial contrast was used to test linear and quadratic effects of graded levels of dietary TABP. Variables with equal variance were subjected to post-hoc analysis using Tukey's honest significance test at 5% level of significance. In the present study, data were presented in the text, and tables are means and standard error of mean (SEM) and statistical significance was estimated at $P < 0.05$ level of significance. The broken line regression model was established in SPSS 20.0. The correlation among growth performance, feed conversion, nutrient utilization, body indices and survival was done in PAST 4.01 software (Hammer et al., 2001).

Table 3

Growth performances, nutrient retention and body indices of rohu, *L. rohita* juveniles.

Variables ¹	Diets ²				SEM	P values		
	TABP0	TABP5	TABP10	TABP15		Overall	Linear	Quadratic
IW (g)	20.53	20.63	20.7	20.43	0.153	0.952	0.886	0.617
FW (g)	40.40 ^b	41.73 ^b	45.73 ^c	36.60 ^a	1.010	< 0.001	0.015	< 0.001
AWG (g)	19.87 ^b	21.10 ^b	25.03 ^c	16.17 ^a	0.993	< 0.001	0.037	< 0.001
FI (g/ fish)	40.27	40.57	40.89	38.62	1.197	0.936	0.716	0.652
PI (g/ fish)	12.16	12.48	12.46	11.93	0.364	0.958	0.853	0.624
FTI (g/ fish)	3.49	3.57	3.59	3.38	0.115	0.944	0.806	0.612
EI (MJ/ fish)	7.30	7.37	7.42	7.00	0.218	0.933	0.714	0.638
BWG (g/day)	0.22	0.23	0.28	0.18	0.011	< 0.001	0.037	< 0.001
WGP (%)	96.79 ^{ab}	102.45 ^{bc}	121.08 ^c	79.22 ^a	4.876	0.001	0.120	0.001
SGR (% /day)	1.13 ^{ab}	1.17 ^{bc}	1.32 ^c	0.97 ^a	0.041	0.001	0.081	0.001
FCR	2.03 ^{ab}	1.91 ^{ab}	1.63 ^a	2.40 ^b	0.096	0.009	0.136	0.005
PER	1.64 ^{ab}	1.71 ^{ab}	2.03 ^b	1.36 ^a	0.084	0.012	0.284	0.007
ANPU (%)	28.38 ^{ab}	29.85 ^{ab}	36.79 ^b	22.10 ^a	1.788	0.006	0.217	0.004
FR (%)	44.05 ^{ab}	48.29 ^b	47.63 ^b	31.57 ^a	2.482	0.026	0.035	0.016
ER (%)	21.05 ^{ab}	21.40 ^{ab}	24.14 ^b	16.61 ^a	0.997	0.026	0.119	0.019
SRP (%)	87.78 ^b	88.11 ^b	97.44 ^c	82.68 ^a	1.332	< 0.001	0.087	< 0.001
HSI (%)	1.88	1.85	2.21	1.48	0.113	0.140	0.349	0.104
VSI (%)	3.67 ^a	3.70 ^a	4.61 ^b	3.07 ^a	0.182	0.002	0.301	0.002
CSI (%)	0.04 ^a	0.05 ^{ab}	0.06 ^b	0.04 ^a	0.002	0.007	0.636	0.003

For growth performance, data are mean values of 3 replicate tanks per treatment (45 fish per replicate tank). For nutrient retention and body indices, data are mean value of 3 replicate tanks per treatment (3 fish per replicate tank). Mean values in the same row with different superscript differ significantly ($P < 0.05$).

¹IW, initial weight; FW, final weight; AWG, average weight gain; FI, feed intake; PI, protein intake; FTI, fat intake; EI, energy Intake; WGP, weight gain percentage; SGR, specific growth rate; FCR, feed conversion ratio; PER, protein efficiency ratio; ANPU, apparent net protein utilization; FR, fat retention; ER, energy retention; SRP, survival percentage; HSI, hepatosomatic index; VSI, viscerosomatic index; CSI, craniosomatic index.

²TABP0, control treatment with 0 g TABP/kg diet; TABP5, treatment with 5 g TABP/kg diet; TABP10, treatment with 10 g TABP/kg diet; TABP15, treatment with 15 g TABP/kg diet.

3. Result

3.1. Chemical composition of TABP

Proximate composition of different ingredients used for feed preparation was illustrated in Table 1. CP, EE, total carbohydrate, neutral detergent fibre, and acid detergent fibre contents of TABP were 35.4, 10.7, 767.7, 310.2, 186.3 and 281.0 g/kg, respectively. Tannin, phytates and saponin contents of TABP were 7.8, 1.8 and 5.1 g/kg, respectively (Table 1).

3.2. Growth performance, nutrient utilization indices, body indices and survival percentage

Dietary TABP improved the average weight gain (AWG) in *L. rohita* fingerlings significantly ($P < 0.05$), exhibited maximum and minimum weight gain (WG) in TABP10 and TABP15, respectively (Table 3); however, no significant difference was reported between TABP0 and TABP5 groups ($P > 0.05$). Growth rates {weight gain percentage (WGP) and SGR}, feed conversion (FCR and PER) and nutrient utilization (ANPU, FR and ER) were significantly affected by dietary TABP in overall as well as quadratically ($P < 0.05$) but not linearly ($P > 0.05$). Highest WGP (121.08%) and SGR (1.32% per day) and PER (2.03) were observed in *L. rohita* fingerlings fed with 10 g/kg TABP among dietary treatments while FCR (1.63) followed a reverse trend (Table 3). Growth rates, feed conversion and nutrient utilization were observed as lowest at 15 g/kg TABP. However the growth rates, feed conversion and nutrient utilization were similar in TABP0 and TABP15 ($P > 0.05$). Feed intake (FI), protein intake (PI) and fat intake (FTI) did not differ in response to dietary TABP ($P > 0.05$). Survival (%) was similar between TABP0 (87.78%) and TABP5 (88.11%) groups ($P > 0.05$) but varied significantly between TABP10 (97.44%) and TABP15 (82.68%) groups ($P < 0.05$). Hepatosomatic index did not differ among treatments ($P > 0.05$). Highest VSI (4.61%), and CSI (0.06%) value was recorded in TABP10 groups compared to all groups ($P < 0.05$).

The correlation matrix of the growth related parameters being represented in Fig. 1. Weight gain percentage (WGP) did not show significant correlation with PER, while FCR found to be negatively correlated ($P < 0.05$). WGP is positively correlated with ANPU, PI, FTI, HSI, CSI, GSI, SRP, ER, EI and SGR. AWG showed a highly negative correlation ($P < 0.05$) with FCR. FCR showed a significantly negative correlation with maximum parameters except HSI, PER and FI.

SGR showed negative highly significant correlation with FCR, non-significant positive correlation with CSI, FR, ANPU, FI and PER where as positive significant correlation with WGP, AWG, PI, FTI, EI, ER, SRP, VSI and HSI. Broken line regression has revealed 12.3 g/kg of TABP as optimum dietary supplement in the feed of *L. rohita* (Fig. 2).

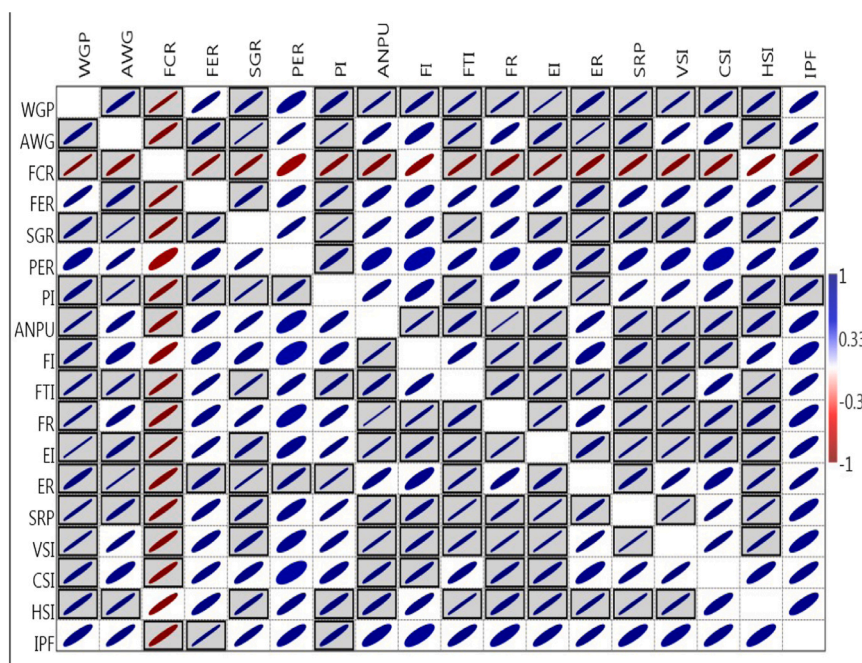


Fig. 1. Correlation matrix of growth performances and nutrient utilization indices of *Labeo rohita*. In the figure the diagonal with boxes indicate the significant at $P < 0.05$. Blue colour is representing positive correlation and red colour negative correlation. The value ranged from -1 to 1 , intermittent colour presents intensity of significance and relative value of correlation. ANPU, apparent net protein utilization; AWG, average weight gain; CSI (%), craniosomatic index; EI, energy Intake; ER, energy retention; FCR, feed conversion ratio; FI, feed intake; FR, fat retention; FTI, fat intake; HSI (%), hepatosomatic index; PER, protein efficiency ratio; PI, protein intake; SGR, specific growth rate; SRP, Survival percentage; VSI (%), viscerosomatic index; WGP, weight gain percentage.

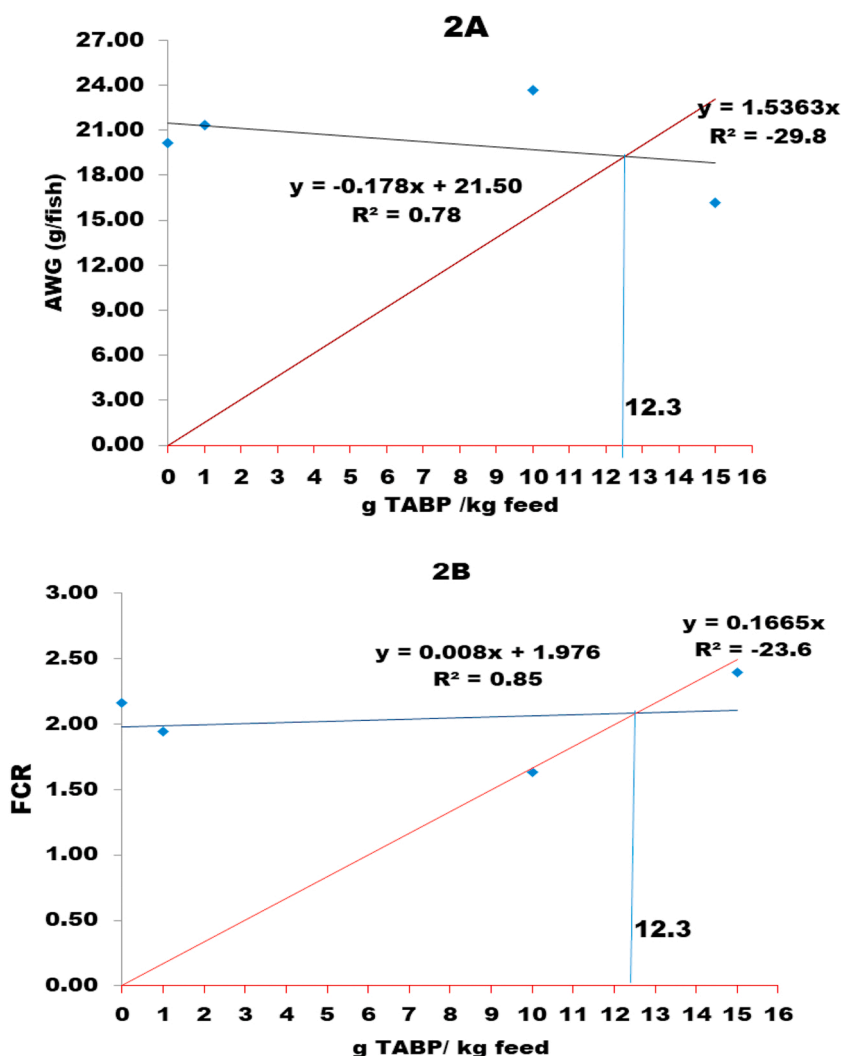


Fig. 2. (A and B) Determination of the optimal dietary level (g/kg) of *Terminalia arjuna* bark powder (TABP) in the feed of Indian major carp *Labeo rohita* by means of broken-line regression model using the average weight gain (AWG, A) and feed conversion ratio (FCR, B).

3.3. Carcass composition

The carcass composition of experimental fish is enumerated in Table 4. The moisture content in *L. rohita* exhibited a linear relation with graded levels of dietary TABP ($P < 0.05$). Highest moisture content was recorded in TABP10 (707.58 g/kg fish) and TABP15 (707.50 g/kg fish), followed by TABP5 (703.59 g/kg fish) and minimum in TABP0 (694.16 g/kg fish). The CP contents varied among the treatment groups linearly as well as overall ($P < 0.05$); categorically TABP0, TABP5 and TABP15 did not differ significantly

Table 4

Carcass composition (wet weight basis) of *L. rohita* juveniles of different experimental groups.

Nutrients (g/kg)	Initial	Diets ¹				SEM	P values		
		TABP0	TABP5	TABP10	TABP15		Overall	Linear	Quadratic
Moisture	717.97 ± 5.06	694.16 ^a	703.59 ^{ab}	707.58 ^b	707.50 ^b	1.985	0.020	0.005	0.104
Crude protein	161.97 ± 1.72	167.49 ^{ab}	168.01 ^{ab}	172.06 ^b	162.99 ^a	1.177	0.023	0.215	0.016
Ether extract	62.09 ± 0.38	69.52 ^{ab}	71.18 ^b	65.16 ^{ab}	63.88 ^a	1.119	0.029	0.018	0.903
Ash	43.31 ± 1.52	45.81	43.34	42.18	41.7	0.703	0.154	0.037	0.436

The data are mean values of 3 replicate tanks per treatment (3 fish per replicate tank). Mean values in the same row with different superscript differ significantly ($P < 0.05$).

¹TABP0, control treatment with 0 g TABP/kg diet; TABP5, treatment with 5 g TABP/kg diet; TABP10, treatment with 10 g TABP/kg diet; TABP15, treatment with 15 g TABP/kg diet.

($P < 0.05$), while TABP10 exhibited higher values than control ($P < 0.05$). The CP of initial fish was 161.97 g/kg fish, and highest CP content was in TABP10 (172.06 g/kg fish) followed by TABP5 (168.01 g/kg fish) and TABP15 (162.99 g/kg fish). EE content in *L. rohita* fingerlings follows an inverse correlation with dietary TABP, exhibiting minimum EE content in TABP15 group. No difference was observed in ash content of *L. rohita* in response to dietary TABP ($P > 0.05$). But, ash content was decreased linearly with graded TABP levels ($P < 0.05$) exhibiting maximum and minimum values in TABP5 and TABP15, respectively.

3.4. Nucleic acid analyses

RNA contents and RNA: DNA ratio of different tissues viz. muscle, liver, kidney, intestine, and brain in *L. rohita* fingerlings were significantly affected by dietary TABP ($P < 0.05$) in overall as well as linearly and quadratically (Table 5). While DNA contents were independent to dietary TABP ($P > 0.05$).

RNA-M showed insignificant difference between TABP0 (0.23 µg/ml) and TABP15 (0.26 µg/ml) and significantly differ with TABP5 (0.38 µg/ml) and TABP10 (0.58 µg/ml). RNA-L showed highly significant difference among treatments groups and values showed trends as follows: TABP15 (0.94 µg/ml) < TABP0 (1.15 µg/ml) < TABP5 (1.22 µg/ml) < TABP10 (1.37 µg/ml). RNA-K, RNA-I and RNA-B also showed highly significant difference among treatment groups ($P > 0.05$) being highest value for TABP10.

Amongst RNA analyses the highest value was observed in treatment TABP10 (1.42 µg/ml) of brain. Irrespective of treatments and across the vital organs the RNA values showed a trend as follows: RNA-B > RNA-L > RNA-K > RNA-I > RNA-M. Similarly, highest RNA: DNA ratio of vital tissues were noticed in TABP10 and minimum in TABP15 group.

4. Discussion

The feeding of herbal product in the feed of livestock and aquaculture has been used and continually increasing due to surging demand of conventional feed ingredients. A balance diet for a commercial culture includes a capsule of ingredient that supplies growth promoter, immunostimulants and basic nutrient to maintain general functions and metabolisms. It has reported that plant-based feed ingredient performances are also at par with their counterparts in terms of replacement of fish meal with local plant sources such as low protein fish meal plus corn gluten meal, wheat gluten meal, soybean meal, nonconventional feed ingredients and now herbal feed ingredient, will not be only beneficial to achieve better growth performance in fish but also cut down production costs, thereby increasing the net profit (Shakya, 2017; Hua et al., 2019). There are two major factors to enhance the benefit ratio for farmer by lowering the operational cost of fish production. First factor explains the replacement of the expensive basic feed ingredients with cheaper, easily available nonconventional feed ingredient (Hassan et al., 2016; Jayant et al., 2018; Yadav et al., 2020). While another factor deals with exploration and screening of herbal feed additives as growth enhancer, and immunostimulant etc. in fish (Robb and Crampton, 2013), which has been substantiated in the present study by escalating the growth performances and enhancing survival and nutrient utilization in *L. rohita* fed with TABP at inclusion level of 10 g/kg feed.

AWG and WGP are the relative function of feed intake (Gonzales and Law, 2013) and varied as per the beneficial moiety of the feed ingredients which in consonance of the present study that showed increased AWG and WGP in TABP10 group and reduction in TABP15 group. It might be due to impaired function at higher dose of herbal material that fostering comparatively less growth performances of fish in terms of AWG and WGP, FCR represent the quality of feed to enhance relative live weight gain. Nugroho et al. (2019) stated that *Pangasianodon hypophthalmus* fed at higher dose (250 g/kg) of *Terminalia catappa* exhibited reduction in growth rates and might be

Table 5
Nucleic acid analyses of vital organs of *L. rohita* juveniles.

Nucleic acid	Tissues	Diets ¹				SEM	P values		
		TABP0	TABP5	TABP10	TABP15		Overall	Linear	Quadratic
RNA (µg/ml)	Muscle	0.23 ^a	0.38 ^b	0.58 ^c	0.26 ^a	0.042	< 0.001	< 0.001	< 0.001
	Liver	1.15 ^b	1.22 ^c	1.37 ^d	0.94 ^a	0.047	< 0.001	< 0.001	< 0.001
	Kidney	1.11 ^b	1.17 ^c	1.26 ^d	0.90 ^a	0.040	< 0.001	< 0.001	< 0.001
	Intestine	1.10 ^b	1.16 ^c	1.19 ^c	0.79 ^a	0.048	< 0.001	< 0.001	< 0.001
	Brain	1.21 ^b	1.23 ^b	1.42 ^c	0.99 ^a	0.047	< 0.001	< 0.001	< 0.000
DNA (µg/ml)	Muscle	0.19	0.23	0.24	0.19	0.017	0.671	0.869	0.221
	Liver	0.95	1.06	1.05	0.94	0.022	0.075	0.830	0.013
	Kidney	0.89	0.99	1.02	0.91	0.025	0.197	0.643	0.047
	Intestine	0.40	0.49	0.52	0.43	0.021	0.129	0.438	0.032
	Brain	0.51	0.56	0.58	0.54	0.018	0.597	0.509	0.267
RNA- DNA ratio	Muscle	1.29 ^a	1.78 ^{ab}	2.48 ^b	1.43 ^{ab}	0.177	0.043	0.361	0.016
	Liver	1.21 ^c	1.15 ^{ab}	1.31 ^c	1.00 ^a	0.037	0.002	0.019	0.008
	Kidney	1.25 ^c	1.19 ^{ab}	1.24 ^c	0.99 ^a	0.037	0.011	0.006	0.083
	Intestine	2.77 ^b	2.40 ^{ab}	2.31 ^{ab}	1.85 ^a	0.118	0.018	0.003	0.801
	Brain	2.39 ^b	2.22 ^{ab}	2.46 ^{ab}	1.85 ^a	0.092	0.049	0.050	0.141

The data are mean values of 3 replicate tanks per treatment (3 fish per replicate tank). Mean values in the same row with different superscript differ significantly ($P < 0.05$).

¹TABP0, control treatment with 0 g TABP/kg diet; TABP5, treatment with 5 g TABP/kg diet; TABP10, treatment with 10 g TABP/kg diet; TABP15, treatment with 15 g TABP/kg diet.

associated with higher concentration of bioactive compounds in fish feed. SGR also showed same trend as most of the other growth performances parameters and almost reverse trend to FCR that is in agreement with previous studies of (Jayasree et al., 2016) on growth promoting effect of Triphala (*Emblica officinalis*, *Terminalia bellerica* and *Terminalia chebula*) biofortification on *Oreochromis mossambicus*. Similarly, corollary to the present study, Asadi et al. (2012) revealed that *Phyllanthus niruri* and *Aloe vera* when supplemented in the diet of *L. rohita*, the *Phyllanthus niruri* fed group resulted higher SGR, might be attributed to improved feed utilization and high rate of protein synthesis. The improvisation in growth performance parameters could be attributed to the credence of active principles of TABP that has been fortified in the diet manifested a growth promoting effect. The active components might have enhanced the feed utilization by the assimilation of most nutrients into flesh. Present results are in agreement with Ahilan et al. (2010) who reported that the inclusion of *Phyllanthus niruri* and *Aloe vera* had enhanced the growth performance of goldfish, *Carassius auratus*. In the present study, the AWG was varied between 79% and 121% for TABP0 and TABP10, respectively at the end of 90 days feeding trial. The growth rates of Indian major carps (IMCs) in unlikely higher like other fish species such as pangas in controlled conditions. The results of the present study are in accordance to the previous research who have reported that *L. rohita* could gain up to 70–100% in 8 weeks feeding trials (Jayant et al., 2020, 2021).

Among nutrient utilization parameters, ANPU, FR, and ER showed significantly higher value in TABP10 group that validate the growth promoting activity and nutrient retention capability of TABP based feed which is in accordance to the anticipation of Hua et al. (2019). Similar FI, PI, FTI and EI in control and treatment groups might be associated to presence of low levels of tannin contents of the experimental diets. As the tannin content of TABP (7.8 g/kg) was lower than the tannin tolerable limits i.e. < 20 g/kg (Francis et al., 2001).

The survival (%) of *L. rohita* was significantly higher in TABP10 as compared to the treatment groups that is attributed to the improved health status upon feeding of herbal feed at a particular dose which is in agreement with previous studies (Sahu et al., 2007a, 2007b, 2008; Al-Thobaiti et al., 2018; Hassan et al., 2018; Gabriel, 2019). The feed ingredient processing is an intermediary process of feed formulation to increase the feed palatability and scalability for shaping it as quality feed (Hajra et al., 2013). Suely et al. (2016) reported that *T. arjuna* bark extract was found to be toxic while applied at 4.71 mg/L in *Heteropneustes fossilis* in the form of immersion. Nugroho et al. (2016) evaluated the effects of *Terminalia catappa* leaves extract on the water quality properties, survival and blood profile of ornamental fish (*Betta* sp.) cultured. Author suggested that immersing leaf extract of *T. catappa* above 375 ppm is beneficial to enhance survival, RBC, WBC, and Hb of *Betta* sp. To substantiate the presumptions of (Hajra et al., 2013) in the present study, TABP has got processed during feed formulation that may lead to reduce the anti-nutritional factor (Tannin) and got improved status of bioactive principle due to application of phytase enzyme, are at accordance to the previous study (Kumar et al., 2012). Meena et al. (2020b) has reported the ethanolic bark extract possess 22% extract yield and, in our study, effective TABP at 10 g TABP/kg feed or 12.3 g/kg of feed that correspondence to 220 mg or 270 mg that is less than the previous studies. In addition, Meena et al. (2020b) has reported that ethanolic bark extract found to be moderately toxic at 1030 mg/L concentration in *Artemia salina* model that will be many folds higher in case of fish, apparently that validate the present study as to be safe for experimental animal. The variation from the previous might be due to method of extraction, solvent systems etc. (Meena et al., 2020a). The anti-nutritional factor, saponin normally does not interfere the normal functioning of the feed physiology and tannin and phytate has wide ranges of their distribution across the plant kingdom (Habtamu and Ratta, 2014) and in the present study these were well within the tolerable range.

The body indices such as VSI (%), and CSI (%) proved the comparative variations in nutrient assimilation of tissue. Present study conform the presumption by proving the gradual increase of all parameters up to TABP10 and then sharp decline in TABP15 group. CSI showed a less significance among the treatments that might be due to the fact that CSI mainly credence to the underlying mechanism upon herbal dietary inclusion unlike other parameters which is in accordance to previous study (Jayant et al., 2018; Prusty et al., 2011). *Terminalia bark* is known for its anti-cholesteric property that has been investigated in animal models (Subramaniam et al., 2011; Kumar et al., 2013; Dwivedi, and Chopra, 2014). Nucleic acid analyses for all vital organs followed the same trend and decline in case of TABP15 treatment that might be attributed to impair functions of protein synthesis machinery by fostering low CSI.

In the present study, highest CP (g/kg) content was recorded in TABP10 group as compared to other groups and lowest value in TABP15 treatment might be due to inefficient protein retention, which is confirmed with the result of Jayant et al. (2018). It has been reported that *T. arjuna* bark has anti-lipidic and anti-cholesteric activity. The EE followed a negative relationship with dietary TABP and minimum EE content was observed at 15 g TABP/kg feed and is in agreement with Subramaniam et al. (2011). Ash (g/kg) content was reported maximum in TABP0 and then decreased linearly as TABP5 > TABP10 > TABP15. This decreasing trend might be attributed to the presence of inorganic soluble upon enhanced herbal dietary supplementation. Shah et al. (2017) documented that ash contents increased during starvation and stressful environment. Moisture content showed reverse trend to EE content, TABP10 and TABP15 being highest moisture content is in agreement to the studies revealing higher moisture content while fed at graded level of unconventional feed ingredient interfering feed intake (Jayant et al., 2018; Arguello-Guevara et al., 2018). Conversely, moisture content has reported to be decreased significantly ($P < 0.05$) in fish groups fed different herbal plants compared with control group (Hassan et al., 2018). This deviation in our study from the above study may likely due to modulatory effects of herbal plant on biochemical composition of fish species. In addition, the Growth rates, feed conversion, and nutrient utilization were varied non-significantly between TABP0 and TABP15 group that might be negative feedback mechanism of the TABP bioactive principle available in bark powder; however, we do not have sufficient fact to validate this hypothesis.

RNA contents and RNA:DNA ratio has been used as an indicator to assess nutritional status of fish (Ciji et al., 2013). In the present study the RNA:DNA ratio followed as TABP0 > TABP15 > TABP5 > TABP10 that indicate that might be due to impaired RNA synthesis parameters in TABP15 and improved function in TABP10 which significantly differ with treatments groups for all the vital organs. RNA:DNA ratio were highest in intestine and lowest in liver which is due to de-activation and activation of RNA synthesis in intestine and liver, respectively that coincides with the results of Ciji et al. (2013).

5. Conclusion

The present study is of its first kind, an indoor feed trial with prospective *T. arjuna* bark powder as feed additive (growth promotor) in the feed of *L. rohita*. Fish fed with 10 g *T. arjuna* bark powder/ kg diet exhibited higher growth rates, feed conversion, nutrient retention and body indices than other fed groups. Based on broken-line regression, the minimal dose of *T. arjuna* bark powder required to support maximum growth performance and feed conversion was 12.3 g/kg feed (independent variables- dietary *T. arjuna* bark powder levels, and dependent variables- feed conversion ratio and average weight gain). As this is an in-vitro bioconversion study which needs to be extrapolated in ex-situ conditions for recommending a package of practice for fishers/fish farmers and other stakeholders.

CRedit authorship contribution statement

D.K. Meena: Investigation, Data curation, Software, Methodology, Formal analysis, Validation, Writing – original draft. **A.K. Sahoo:** Conceptualization, Software, Writing – review & editing. **M. Jayant:** Formal analysis, article proof read. **N.P. Sahu:** Conceptualization, Supervision, Writing – review & editing. **P.P. Srivastava:** Conceptualization, Supervision, Validation. **H.S. Swain:** Experimental facility. **B.K. Behera:** Conceptualization, proof read. **K. Satvik:** Software, wet lab assistance. **B.K. Das:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Conflicts of interest

We declare no conflicts of interest. Each has contributed to the article and it is an original research neither published before nor under consideration for publication.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.anifeedsci.2021.115132](https://doi.org/10.1016/j.anifeedsci.2021.115132).

References

- Ahilan, B., Nithiya, P.A., Ravaneshwaran, K., 2010. Influence of certain herbal additives on the growth, survival and disease resistance of goldfish, *Carassius auratus* (Linnaeus). *Tamilnadu. J. Vet. Anim. Sci.* 6 (1), 5–11.
- Al-Thobaiti, A., Al-Ghanim, K., Ahmed, Z., Suliman, E.M., Mahboob, S., 2018. Impact of replacing fishmeal by a mixture of different plant protein sources on the growth performance in Nile Tilapia (*Oreochromis niloticus* L.) diets. *Braz. J. Biol.* 78 (3), 525–534. <https://doi.org/10.1590/1519-6984.172230>.
- Amalraj, A., Gopi, S., 2016. Medicinal properties of Terminalia arjuna (Roxb.) Wight & Arn.: A review. *J. Tradit. Complement. Med.* 7 (1), 65–78. <https://doi.org/10.1016/j.jtcm.2016.02.003>.
- AOAC, 1995. Official Methods of Analysis of AOAC International, sixteenth ed. AOAC International, Arlington, VA, USA, p. 1995.
- APHA, 1995. Standard Methods for the Examination of Water and Wastewater, eighteenth ed. American Public Health Association, Washington, USA, p. 1193.
- Arguello-Guevara, W., Apolinario, W., Bohorquez-Cruz, M., Reinoso, S., Rodriguez, S., Sonnenholzner, S., 2018. Effects of intermittent feeding on water quality, skin parasites, feed consumption, and growth performance of juvenile longfin yellowtail *Seriola rivoliana* (Valenciennes, 1833). *Aquac. Res.* 49 (11), 3586–3594. <https://doi.org/10.1111/are.13825>.
- Asadi, M.S., Mirvaghefi, A.R., Nematollahi, M.A., Banaee, M., Ahmadi, K., 2012. Effects of Watercress (*Nasturtium nasturtium*) extract on selected immunological parameters of rainbow trout (*Oncorhynchus mykiss*). *Open Vet. J.* 2 (1), 32–39.
- Brett, J.R., 1973. Energy expenditure of sockeye salmon, *Oncorhynchus nerka*, during sustained performance. *J. Fish. Res. Board Can.* 30, 1799–1809. <https://doi.org/10.1139/f73-290>.
- Ciji, A., Sahu, N.P., Pal, A.K., Akhtar, M.S., Viji, P., 2013. Effect of dietary gelatinized starch level and rearing temperature on fatty acid profile and DNA/RNA ratio of *Labeo rohita* (Hamilton) fingerlings. *Isr. J. Aquac. -Bamid.* 64, 836–841.
- Darambazar, E., 2018. Method of phytic acid determination in food, feed, and plant materials. *Assay protocol*, University of Saskatchewan, Canada, p. 4, doi: 10.13140/RG.2.2.14852.27520.
- De, B.C., Meena, D.K., Behera, B.K., Das, P., Mohapatra, P.K.D., Sharma, A.P., 2014. Probiotics in fish and shellfish culture: immunomodulatory and ecophysiological responses. *Fish. Physiol. Biochem.* 40 (3), 921–971. <https://doi.org/10.1007/s10695-013-9897-0>.
- Dutta, S., 2015. Aquaponics: its present status and potential. *Fish. Chimes* 43 (11), 1–4.
- Dwivedi, S., Chopra, D., 2014. Revisiting Terminalia arjuna – an ancient cardiovascular drug. *J. Tradit. Complement. Med.* 4 (4), 224–231. <https://doi.org/10.4103/2225-4110.139103>.
- FAO, 2016. Contributing to foodsecurity and nutrition for all, Rome: Food and Agriculture Organization, 2016, pp. 200.
- FAO, 2020. The State of World Fisheries and Aquaculture – Sustainability in action. FAO Fisheries and Aquaculture Department, Rome, Italy, pp. 2–75. <https://doi.org/10.4060/ca9229en>.
- Francis, G.M., Becker, K., Makkar, H.P.S., 2001. Anti-nutritional factors present in plant derived alternate fish feed ingredients and their effects in fish. *Aquaculture* 199 (3–4), 197–227. [https://doi.org/10.1016/S0044-8486\(01\)00526-9](https://doi.org/10.1016/S0044-8486(01)00526-9).
- Gabriel, N.N., 2019. Review on the progress in the role of herbal extracts in tilapia culture. *Cogent Food Agric.* 5 (1), 1619651 <https://doi.org/10.1080/23311932.2019.1619651>.
- Gonzales Jr, J.M., Law, S.H.W., 2013. Feed and feeding regime affect growth rate and gonadosomatic index of adult zebrafish (*Danio rerio*). *Zebrafish* 10 (4), 532–540. <https://doi.org/10.1089/zeb.2013.0891>.
- Hajra, A., Mazumder, A., Verma, A., Ganguly, D.P., Mohanty, B.P., Sharma, A.P., 2013. Antinutritional factors in plant origin fish feed ingredients: the problems and probable remedies. *Adv. Fish. Res.* 5, 193–202.
- Hammer, O., Harper, D.A.T., Ryan, P.D., 2001. PAST: paleontological statistics software package for education and data analysis. *Palaeontol. Electron.* 4 (1), 9.
- Harborne, J.B., 2005. *Phytochemical Methods – A Guide to Techniques of Plant Analysis*, third ed. Springer Ltd., Netherlands, p. 302.

- Hassan, A.A.M., Yacout, M.H., Khalel, M.S., Hafsa, S.H.A., Ibrahim, M.A.R., Mocuta, D.N., Dedi, L., 2018. Effects of some herbal plant supplements on growth performance and the immune response in Nile Tilapia (*Oreochromis niloticus*). In: "Agriculture for Life, Life for Agriculture" Conference Proceedings, Bucharest, Romania, Sciend, Vol. 1, No. 1, pp. 134–141. <https://doi.org/10.2478/alife-2018-0020>.
- Hassan, M.A., Aftabuddin, M., Meena, D.K., Mishal, P., Gupta, S.D., 2016. Effective utilization of distiller's grain soluble—an agro-industrial waste in the feed of cage-reared minor carp *Labeo bata* in a tropical reservoir, India. *Environ. Sci. Pollut. Res.* 23, 1–6. <https://doi.org/10.1007/s11356-016-6732-z>.
- Hassan, M.A., Aftabuddin, M., Meena, D.K., Saha, S., Dasgupta, S., Sharma, A.P., 2014. Effects of black thorn, *Kaempferia galanga* single or in combination with yeast probiotic on feed palatability, growth performance and product quality of *Labeo rohita* fingerling (Hamilton). *Turk. J. Fish. Aquat. Sci.* 14, 1–2. https://doi.org/10.4194/1303-2712-v14_4_10.
- Habtmu, G.F., Ratta, N., 2014. Antinutritional factors in plant foods: potential health benefits and adverse effects. *Int. J. Food Sci. Nutr.* 3 (4), 284–289. <https://doi.org/10.11648/j.ijnfs.20140304.18>.
- Hua, K., Cobcroft, J.M., Cole, A., Condon, K., Jerry, D.R., Mangott, A., Strugnell, J.M., 2019. The future of aquatic protein: implications for protein sources in aquaculture diets. *One Earth* 1 (3), 316–329. <https://doi.org/10.1016/j.oneear.2019.10.018>.
- Huynh, T.G., Nguyen, T.P., Vu, N.U., Jack, M., Truong, Q.P., 2016. Assessment of water quality in catfish (*Pangasianodon hypophthalmus*) production systems in the Mekong delta. *Can Tho Univ. J. Sci.* 3, 71–78. <https://doi.org/10.22144/ctu.jen.2016.107>.
- IBM Corp. Released, 2011. *IBM SPSS Statistics for Windows, Version 20.0*. IBM Corp., Armonk, NY, USA.
- Jayant, M., Hassan, M.A., Srivastava, P.P., Meena, D.K., Kumar, P., Kumar, A., Wagde, M.S., 2018. Brewer's spent grains (BSGs) as feedstuff for striped catfish, *Pangasianodon hypophthalmus* fingerlings: an approach to transform waste into wealth. *J. Clean. Prod.* 199, 716–722. <https://doi.org/10.1016/j.jclepro.2018.07.213>.
- Jayant, M., Sahu, N.P., Deo, A.D., Garg, C.K., Yadav, R., Gupta, S., 2021. Effective valorization of agro-waste of castor oil extraction industry as feedstock for sustainable fish production. *Biofuels, Bioprod. Bioref.* 15, 1126–1140. <https://doi.org/10.1002/bbb.2228>.
- Jayant, M., Sahu, N.P., Deo, A.D., Subodh, G., Garg, C.K., Valappil, R.K., 2020. Nutritional evaluation of fermented sweet potato leaf meal as a replacer of de-oiled rice bran in the diet of *Labeo rohita* fingerlings. *J. Exp. Zool. India* 23 (1), 61–70.
- Jayasree, S., Usha, S., Thomas, A.A., 2016. Growth promotion effect of triphala biofortification on *Oreochromis mossambicus*. *Int. J. Sci. Res.* 5 (9), 1126–1130. <https://doi.org/10.21275/ART20161619>.
- Kumar, C., Kumar, R., Nehar, S., 2013. Phytochemical properties, total antioxidant status of acetone and methanol extract of *Terminalia arjuna* Roxb. bark and its hypoglycemic effect on Type-II diabetic albino rats. *J. Pharmacogn. Phytochem.* 2 (1), 199–208.
- Kumar, V., Sinha, A.K., Makkar, H.P., De Boeck, G., Becker, K., 2012. Phytate and phytase in fish nutrition. *J. Anim. Physiol. Anim. Nutr.* 96 (3), 335–364. <https://doi.org/10.1111/j.1439-0396.2011.01169.x>.
- Makkar, H.P., Blümmel, M., Borowy, N.K., Becker, K., 1993. Gravimetric determination of tannins and their correlations with chemical and protein precipitation methods. *J. Sci. Food Agric.* 61, 161–165. <https://doi.org/10.1002/jsfa.2740610205>.
- Meena, D.K., Sahoo, A.K., Chowdhury, H., Swain, H.S., Sahu, N.P., Behera, B.K., Das, B.K., 2020a. Effects of extraction methods and solvent systems on extract yield, proximate composition and mineral profiling of *Terminalia arjuna* (Arjuna) dry powders and solvent extracts. *J. Innov. Pharm. Biol. Sci.* 7 (1), 22–31.
- Meena, D.K., Sahoo, A.K., Srivastava, P.P., Sahu, N.P., Jadhav, M., Gandhi, M., Swain, H.S., Borah, S., Das, B.K., 2021a. On valorization of solvent extracts of *Terminalia arjuna* (arjuna) using DNA scission and free radical scavenging improves coupling responses and cognitive functions under in vitro conditions. *Sci. Rep.* 11, 10656. <https://doi.org/10.1038/s41598-021-88710-w>.
- Meena, D.K., Sahoo, A.K., Swain, H.S., Borah, S., Srivastava, P.P., Sahu, N.P., Das, B.K., 2020b. Prospects and perspectives of virtual in-vitro toxicity studies on herbal extracts of *Terminalia arjuna* with enhanced stratagem in Artemia salina model: A panacea to explicit the credence of solvent system in brine shrimp lethality bioassay. *Emir. J. Food Agric.* 32 (1), 25–37. <https://doi.org/10.9755/efja.2020.v32.i1.2055>.
- Meena, D.K., Sahu, N.P., Srivastava, P.P., Jadhav, M., Prasad, R., Malick, R.C., Sahoo, A.K., Behera, B.K., Mohanty, D., Das, B.K., 2021b. Effective valorisation of facile extract matrix of *Terminalia arjuna* (Roxb) against elite microbes of aquaculture industry – a credence to bioactive principles: can it be a sustainability paradigm in designing broad spectrum antimicrobials? *Ind. Crops Prod.* 171, 113905. <https://doi.org/10.1016/j.indcrop.2021.113905>.
- Nugroho, R.A., Aryani, R., Anggraini, W.D.C., Hardi, E.H., Rudianto, R., Kusumawati, E., Manurung, H., 2019. Dietary *Terminalia catappa* leaves reduced growth performance but increased hematological profiles and survival rate of *Pangasianodon hypophthalmus*. *IOP Conf. Ser.: Earth. Environ. Sci.*, 348, 012033. doi: 10.1088/1755-1315/348/1/012033.
- Nugroho, R.A., Manurung, H., Saraswati, D., Ladyescha, D., Nur, F.M., 2016. The effects of *Terminalia catappa* L. leaves extract on the water quality properties, survival and blood profile of ornamental fish (*Betta* sp) cultured. *Biosaintifika: J. Biol. Educ.* 8 (2), 240–247. <https://doi.org/10.15294/biosaintifika.v8i2.6519>.
- Prusty, A.K., Kohli, M.P.S., Sahu, N.P., Pal, A.K., Saharan, N., Mohapatra, S., Gupta, S.K., 2011. Effect of short-term exposure of fenvalerate on biochemical and haematological responses in *Labeo rohita* (Hamilton) fingerlings. *Pestic. Biochem. Physiol.* 100 (2), 124–129. <https://doi.org/10.1016/j.pestbp.2011.02.010>.
- Rao, Y.V., Das, B.K., Jyotirmayee, P., Chakrabarti, R., 2006. Effect of *Achyranthes aspera* on the immunity and survival of *Labeo rohita* infected with *Aeromonas hydrophila*. *Fish Shellfish Immunol.* 20 (3), 263–273. <https://doi.org/10.1016/j.fsi.2005.04.006>.
- Robb, D.H.F., Crampton, V.O., 2013. On-farm feeding and feed management: perspectives from the fish feed industry. *On-farm feeding and feed management in aquaculture*. In: Hasan, M.R., New, M.B. (Eds.), *FAO Fisheries and Aquaculture Technical Paper No. 583*. FAO, Rome, Italy, pp. 489–518.
- Sahu, S., Das, B.K., Mishra, B.K., Pradhan, J., Samal, S.K., Sarangi, N., 2008. Effect of dietary *Curcuma longa* on enzymatic and immunological profiles of rohu, *Labeo rohita* (Ham.), infected with *Aeromonas hydrophila*. *Aquac. Res.* 39 (16), 1720–1730. <https://doi.org/10.1111/j.1365-2109.2008.02048.x>.
- Sahu, S., Das, B.K., Mishra, B.K., Pradhan, J., Sarangi, N., 2007a. Effect of *Allium sativum* on the immunity and survival of *Labeo rohita* infected with *Aeromonas hydrophila*. *J. Appl. Ichthyol.* 23 (1), 80–86. <https://doi.org/10.1111/j.1439-0426.2006.00785.x>.
- Sahu, S., Das, B.K., Pradhan, J., Mohapatra, B.C., Mishra, B.K., Sarangi, N., 2007b. Effect of *Magnifera indica* kernel as a feed additive on immunity and resistance to *Aeromonas hydrophila* in *Labeo rohita* fingerlings. *Fish Shellfish Immunol.* 23 (1), 109–118. <https://doi.org/10.1016/j.fsi.2006.09.009>.
- Schneider, W.C., 1957. Determination of nucleic acids in tissue by pentose analysis. *Methods Enzymol.* 3, 680–684. [https://doi.org/10.1016/S0076-6879\(57\)03442-4](https://doi.org/10.1016/S0076-6879(57)03442-4).
- Shah, S.Q.A., Hussain, M.Z., Ali, M., Salam, A., 2017. Effect of stress conditions on body composition parameters of farmed rohu (*Labeo rohita*). *Turkish. J. Fish. Aquat. Sci.* 17 (3), 471–476. https://doi.org/10.4194/1303-2712-v17_3_03.
- Shakya, S.R., 2017. Effect of herbs and herbal products feed supplements on growth in fishes: a review. *Nepal J. Biotechnol.* 5 (1), 58–63. <https://doi.org/10.3126/njb.v5i1.18870>.
- Subramaniam, S., Subramaniam, R., Rajapandian, S., Uthrapathi, S., Gnanamanickam, V.R., Dubey, G.P., 2011. Anti-atherogenic activity of ethanolic fraction of *Terminalia arjuna* bark on hypercholesterolemic rabbits. *Evid. Based Complement. Altern. Med.* 2011, 487916. <https://doi.org/10.1093/ecam/nek003> (1–8).
- Suely, A., Zabeed, H., Ahmed, A.B.A., Mohamad, J., Nasiruddin, M., Sahu, J.N., Ganesan, P., 2016. Toxicological and hematological effect of *Terminalia arjuna* bark extract on a freshwater catfish, *Heteropneustes fossilis*. *Fish Physiol. Biochem.* 42 (2), 431–444. <https://doi.org/10.1007/s10695-015-0149-3>.
- Van Soest, P.J., Robertson, J.B., Lewis, B.A., 1991. Methods for dietary fibre, neutral detergent fibre and non-starch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74, 3583–3597. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2).
- Yadav, G., Meena, D.K., Sahoo, A.K., Das, B.K., Sen, R., 2020. Effective valorization of microalgal biomass for the production of nutritional fish-feed supplements. *J. Clean. Prod.* 243, 118697. <https://doi.org/10.1016/j.jclepro.2019.118697>.